

	FIR VEHICLE TESTING		30/10/2023
FIR No :FIR/Testing-Outdoor/518300/2023/6034			
Part No/ Group :51832155R			
Subject :Performance evaluation of Signa 2820.T CNG BSVI RDE in comparison with Signa 2818.T CNG OBD 1 base vehicle.			
Test Vehicle	Project Name :Signa 2820.T CNG BSVI RDE Model Name : Signa 2820.T CNG BSVI RDE		Enclosures : Completed FIR
	Chasis Number :PC82553032RL10006 Engine Number :HXX814549		
FIR Status :	Completed		

Role	Name	Signature
Prepared By	Debabrata Das	 30-Oct-2023
Checked By	Swapnil Jadhav	 30-Oct-2023
Approved By	Ajaykumar Jadhav	 30-Oct-2023

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

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Documents : Please click here

[FIR_Testing_Outdoor_51830..](#)

 FIR/Testing-Outdoor/518300/2023/6034		FIR VEHICLE TESTING		Date: 30/10/2023		
Part No / Group: 51832155R		Specification no :- " 259051830006 "			Action by	Failure class

FAILURES CLASS:

- A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.
- B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.
- C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.
- D: Customer perception issues and minor irritants.

3.0 Test Vehicle:

1. Signa 2820.T CNG BSVI RDE (EJ00789)
Chassis No. – PC82553032RL10006
Engine No. – HXX814549
2. Signa 2818.T CNG OBD 1
Chassis No – PC82559012RJ00698
Engine No – FYX802243

Enclosures:

- Annexure 1: Panel chart
Annexure 2: PAT Sheet
Annexure 3: PAT Summary
Annexure 4: Test request
Annexure 5: Test plan

Input Doc Ref. : Testing-Outdoor/15522

DID No : NA

DDP No : NA

Work Order No. :- P.21.EA.458

Make :- Tata Motors Ltd

T I No. – 43364

Earlier Ref: -

- FIR/Testing-Outdoor/515000/2022/4931_Report on performance evaluation of Signa 2818.T 5.7 SGI CNG BSVI in comparison with Signa 2818.T Ashwamedh BSVI vehicle.

4.0 Project Status: DR3
5.0 Introduction:

- As a part of RDE program, Signa 2820.T CNG BSVI RDE vehicle was offered for performance evaluation, and Signa 2818.T CNG BSVI vehicle was taken as Base for performance comparison. This report will reveal detail of this trial with the status.

FIR Status

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 FIR/Testing-Outdoor/518300/2023/6034		FIR VEHICLE TESTING	Date: 30/10/2023	
Part No / Group: 51832155R	Specification no :- " 259051830006 "		Action by	Failure class

6.0 Testing

➤ Following test was conducted during trials.

Sr. No.	Test Description	Standard Referred	Instrument used
1	Acceleration	TST_VT_WG_0002	VBOX SX with GPS, Weighbridge (CRS Gate)
2	WOT Drivability	TST_VT_WG_0002	
3	Bath -Tub Drivability	TST_VT_WG_0261	
4	Gradability	TST_VT_WG_11	
5	Restart Gradability	TST_VT_WG_127	
6	Coastdown	TST_VT_WG_36	

7.0 Vehicle details:

Sr. No.	Parameters	Signa 2818.T CNG	Signa 2820.T CNG
1	Engine	TATA 697 5.7L SGI CNG	TATA 697 5.7L SGI CNG
1.1	No of cylinders	6	6
1.2	Engine Max. Power	176 HP @ 2250 rpm	204 HP @ 2250 rpm
1.3	Engine Max. Torque	650 Nm @ 1500 rpm	700 Nm @ 1500 rpm
1.4	Power to weight (HP/Ton)	6.29	7.28
1.5	Torque to weight (Nm/Ton)	23.21	25
1.6	Engine Calibration Number	57SGITCICBS6	--
1.7	Engine Calibration Maturity	Develop Cal	Develop Cal
1.8	Emission Calibration Maturity	Develop	Develop
1.9	Regeneration Strategy	--	--
1.9.1	Exhaust Pipe Insulation	--	--
2	Fuel Economy Improvement Technologies		
2.1	Engine Cooling Fan	E-Viscous (24" Fan)	E-Viscous (24" Fan)
2.2	Fuel Economy Switch	2 mode (EP)	2 mode (EP)
3	Fuel Economy Improvement Technologies		
3.1	Fuel Tank type	Metal Cylinder	Metal Cylinder
3.2	Driveline & Ratios	G750 6S DD, FGR- 9.19	G750 6S DD, FGR- 9.19
3.3	Gear Ratio	1st: 9.19, 2nd: 4.98, 3rd: 2.97, 4th: 1.93, 5th: 1.39, 6th: 1, Rev: 8.36	1st: 9.19, 2nd: 4.98, 3rd: 2.97, 4th: 1.93, 5th: 1.39, 6th: 1, Rev: 8.36
3.4	RAR	6.43	6.43
3.5	Tire size	295/90R20	295/90R20
4	EOL Parameters		
4.1	LBSC, VAM, GDP	NA	NA
4.2	RSG	74	74

FIR Status

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FIR/Testing-Outdoor/518300/2023/6034

FIR

VEHICLE TESTING

Date: 30/10/2023

Part No / Group: 51832155R

Specification no :- " 259051830006 "

Action
byFailure
class**8.0 Torque table:-****➤ Power mode – Test Vehicle**

float y \ x	800	1000	1200	1400	1600	1800	2000	2200
-1	445	545	680	700	700	690	670	630
0	445	545	680	700	700	690	670	630
1	445	545	680	700	700	690	670	630
2	445	545	680	700	700	690	670	630
3	445	545	680	700	700	690	670	630
4	445	545	680	700	700	690	670	630
5	445	545	680	700	700	690	670	630
6	445	545	680	700	700	690	670	630

➤ Light mode – Test Vehicle

float y \ x	800	1000	1200	1400	1600	1800	2000	2200
-1	445	545	680	700	700	690	670	630
0	445	545	680	700	700	690	670	630
1	431	530	670	701	701	683	663	630
2	431	530	670	701	701	683	663	630
3	445	530	540	540	491	442	442	442
4	445	530	503	503	458	412	412	412
5	445	410	410	410	410	365	365	365
6	445	432	432	432	432	384	384	384

➤ 9.0 Observation:-

Overall performance Signa 2820.T CNG BSVI RDE is better as compared to Signa 2818.T CNG BSVI and PALS position achieved "C" against "UC".

10. Future testing:- As per further development.

FIR Status

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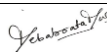
**Annexure - 1
PANEL chart**

Level 1 Attribute:						PAT- Performance and Drivability Specification: Max Power : 204 hp @ 2250 rpm Max Torque : 700 Nm 1500 rpm Driveline : G750 DD, 6.43 RAR , Bridgestone 295/90R20 EOL: LBSC - NA, VAM - NA, GDP - NA, RSG – 74 Kmph						VE CVBU/ PD/00											
Programme / MY : Signa 2820.T CNG RDE BSVI			Gateway : DR3		Owner: Ajaykumar Jadhav		Project Name: Signa 2820.T Cx CNG BSVI RDE			Date - 30/10/2023													
Level 1 Summary:						Statement of Intent (SOI) or DNA Description:																	
<table border="1"> <tr> <td rowspan="2">PALS</td> <td rowspan="2">Level 1:</td> <td>PALS</td> <td colspan="2">Status</td> </tr> <tr> <td>Current/ Gateway</td> <td>Job#1 Predicted</td> </tr> <tr> <td></td> <td></td> <td>UC</td> <td>C</td> <td></td> </tr> </table>						PALS	Level 1:	PALS	Status		Current/ Gateway	Job#1 Predicted			UC	C		Performance: Haulage application Fuel Economy: Haulage application					
PALS	Level 1:	PALS	Status																				
		Current/ Gateway	Job#1 Predicted																				
		UC	C																				

Attribute Target Summary:													
4 Rows	Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	Current Target w.r.t Balanced Target	Status Current/Gateway Signa 2820.T Cx CNG RDE BSVI	Base Vehicle Signa 2818.T CNG BSVI	Competitor Data BS VI Core Benchmark Adjacent Benchmark		Remarks
									Heavy mode	Heavy mode	--	--	
	Specification								LBSC - NA VAM- NA GDP- NA RSG - 74 kmph	LBSC - NA VAM- NA GDP- NA RSG - 74 kmph	--	--	--
	Vehicle GVW (Payload)						kg	28000		28000	--	--	
	Power (Max) @ rpm						HP	204 hp @ 2250 rpm		180 hp @ 2300 rpm	--	--	
	Power to Weight ratio						HP / Ton	7.29		6.43	--	--	
	Torque (Max) @ rpm						Nm	700 Nm @ 1500 rpm		650 Nm @ 1500 rpm	--	--	
	Torque to Weight ratio						Nm / Ton	25.00		23.21	--	--	--
	Tyres size							Bridgestone 295/90 R20		Apollo 295/90 R20	--	--	
	Gear Box Ratios							G750 DD, FGR -9.19		G750 DD, FGR -9.19	--	--	
	Final Drive Ratio							6.43		6.43	--	--	
	Acceleration at rated load					UC		C					
	0-60 kmph						Sec	≤ 65.26	54.23	65.26	--	--	
	0-1000 m						Sec	≤ 83.94	77.12	83.94	--	--	
	Bathtub Drivability (Time) at Rated load					UC		C					
	3rd Gear												
	15 - 20 Kmph						sec	≤ 3.20	2.11	2.46	--	--	
	20 - 25 Kmph						sec	≤ 2.47	2.46	2.45	--	--	
	4th Gear												
	20 - 25 Kmph						sec	≤ 5.94	3.19	3.25	--	--	
	25 - 30 Kmph						sec	≤ 4.17	3.29	3.61	--	--	
	30 - 35 Kmph						sec	≤ 3.70	3.60	3.70	--	--	
	35 - 40 kmph						sec	≤ 5.52	6.01	5.52	--	--	
	5th Gear												
	30 - 40 Kmph						sec	≤ 12.11	9.73	10.13	--	--	
	40 - 50 Kmph						sec	≤ 13.83	10.93	13.83	--	--	
	6th Gear												
	30-40 Kmph						sec	≤ 29.15	14.80	24.36	--	--	
	40-50 Kmph						sec	≤ 16.58	14.37	15.89	--	--	
	50-60 Kmph						sec	≤ 18.80	16.30	18.80	--	--	
	Gradeability at rated load					UC		C					
	Sustain Gradeability						%	≥ 26.38	*29.51	27.58	--	--	Sustain & Restart gradeability Extrapolated from Signa 2818.T CNG BSVI vehicle
	Restart Gradeability						%	≥ 18.00	*23.11	21.60	--	--	

Executive Summary:	1. Target has been taken from Signa 2820.T CNG RDE BSVI Target book 2. Signa 2820.T CNG RDE BSVI was having 95% engine calibration maturation. 3.Target has been taken at-par as per Cx variant approach 4.Vehicle is meeting PALS target except bathtub driveability at 4th gear 35 - 40 Kmph, 5.Sustain & Restart gradeability Extrapolated from Signa 2818.T CNG BSVI vehicle
Tested by:	
 Debabrata Das / Swapnil Jadhav PAT Engineer	
Verified and Approved by:	
 Ajaykumar Jadhav / Sanjeev Kumar PAT Lead / Project Lead	

TATA		TATA MOTORS LTD. ERC,PUNE		PAT ATTRIBUTES SHEET Performance and Drivability Vehicle Level Engine Calibration Features / End of Line OBDII and Diagnostics										Document Version: VE CVBU/PD/00 Document Revision No.: VE CVBU/ M&HCV/ Signa 2820.T CNG Cx RDE/ 00 Date: 30/10/2023								
Department Name: Vehicle Engineering										Project Name: Signa2820.T CNG BSVI RDE												
Sr.No.	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Unit	Test Type	Classification (GDR)	Ref.Standard / Work Guideline	Simulation	Base Vehicle Signa 2818.T CNG BSVI G750 DD 6S RAR 6.43	Core Benchmark	Target	Test Vehicle Signa 2020.T CNG RDE BSVI G750DD RAR - 6.43	Status (RYG)	Remarks	CoC Remark				
A	Performance Evaluation						Heading															
11	Minimum speed						Heading															
11.1	1st gear						Kmph	Standard	Specification	TST/VTW/G02		LBSC - NA VAM- NA GDR- NA RSG - 74 kmph	--	--	LBSC - NA VAM- NA GDR- NA RSG - 74 kmph	--	--					
11.2	2nd gear						Kmph					Heavy Mode	--	--	Heavy Mode	--						
11.3	3rd gear						Kmph					2.1	--	--	2.4	--						
11.4	4th gear						Kmph					4.2	--	--	4.5	--						
11.5	5th gear						Kmph					6.7	--	--	7.6	--						
11.6	6th gear						Kmph					10.2	--	--	11.5	--						
11.7	7th gear						Kmph					15.4	--	--	15.2	--						
11.8	8th gear						Kmph					19.1	--	--	21.3	--						
11.9	Vehicle maximum speed						Kmph					NA	--	--	NA	--						
12	Maximum speed						Heading															
12.1	1st gear						Kmph	Standard	Specification	TST/VTW/G02		9.7	--	--	9.7	--						
12.2	2nd gear						Kmph					16.7	--	--	16.5	--						
12.3	3rd gear						Kmph					27.2	--	--	27.0	--						
12.4	4th gear						Kmph					41.2	--	--	40.7	--						
12.5	5th gear						Kmph					55.9	--	--	54.6	--						
12.6	6th gear						Kmph					74.8	--	--	76.0	--						
12.7	7th gear						Kmph					NA	--	--	NA	--						
12.8	8th gear						Kmph					NA	--	--	NA	--						
12.9	Vehicle maximum speed						Kmph					75.0	--	--	75	--						
14	Acceleration						Heading															
14.2	AC Off						Heading															
14.2.1	0-60 kmph (0-37 mph)						Sec	Standard	Target	TST/VTW/G02		66.26	--	5 66.26	54.23	10						
14.2.4	0-1000 m						Sec					82.94	--	5 83.94	77.12	10						
15	Grade ability						Heading															
15.2	AC Off						Heading															
15.2.1	Maximum (sustained) Gradeability (forward)						%	Standard	Target	TST/VTW/G011		27.58	--	5 26.38	29.51	10	Extrapolated from 2818.T CNG BSVI					
15.2.2	Max Restart Gradeability (forward)						%	Standard	Target	TST/VTW/G127		21.6	--	5 18.00	23.11	10						
16	Coast down (a+bv³)						Heading															
16.2	Laden condition						Heading															
16.2.1	Power consumed at 40 kmph						kW	Standard	Data Generation	TST/VTW/G06		35.6	--	--	--	NA						
16.2.2	a						N					1868.70	--	--	--	NA						
16.2.3	b						Nkmph²					0.34	--	--	--	NA						
16.2.4	Coeff of rolling resistance						--					0.0055	--	--	--	NA						
B	Drivability Evaluation						Heading															
13	WOT Drivability (AC Off)						Heading															
13.1	1st gear 4-6 Km/h						Sec	Standard	Data Generation	TST/VTW/G02		0.65	--	--	0.82	--						
13.2	2nd gear 8-12 Km/h						Sec					1.35	--	--	1.47	--						
13.3	3rd gear 12-20 Km/h						Sec					3.83	--	--	3.78	--						
13.4	4th gear 19-31 Km/h						Sec					9.52	--	--	7.92	--						
13.5	5th gear 25-55 Km/h						Sec					42.02	--	--	35.28	--						
13.6	6th gear 36-60 kmph						Sec					47.89	--	--	35.60	--						
13.10	AC Off (Rate of acceleration)						Heading					Specification					--					
13.11	1st gear 4-6 Km/h						Kmph/sec		Data Generation			3.08	--	--	2.44	--						
13.12	2nd gear 8-12 Km/h						Kmph/sec					2.96	--	--	2.72	--						
13.13	3rd gear 12-20 Km/h						Kmph/sec					2.59	--	--	2.12	--						
13.14	4th gear 19-31 Km/h						Kmph/sec					1.47	--	--	1.52	--						
13.15	5th gear 25-55 Km/h						Kmph/sec					0.36	--	--	0.95	--						
13.16	6th gear 36-60 kmph						Kmph/sec		0.29			--	--	0.67	--							
14	Rattub Curve - Part Throttle Drivability (AC Off)						Heading															
14.7	2nd gear						Heading															
14.13	3rd gear						Heading															
14.14	15-20 kmph						sec	Standard	Target Balanced	TST/VTW/G01		2.46	--	5 3.39	2.11	10						
14.15	20 - 25 kmph						sec					2.45	--	5 2.47	2.46	10						
14.1	4th gear						Heading					Specification					--					
14.4	20-25 kmph						sec					3.25	--	5 5.94	3.19	10						
14.5	25-30 kmph						sec					3.61	--	5 4.17	3.29	10						
14.6	30-35 kmph						sec					3.70	--	5 3.79	3.60	10						
14.7	35-40 kmph						sec					5.52	--	5 5.62	6.01	10						
14.8	5th gear						Heading					Specification					--					
14.9	30 - 40 Km/h						sec		Target Balanced			10.13	--	5 12.11	9.73	10						
14.10	40 - 50 Km/h						sec					13.83	--	5 13.83	10.93	10						
14.13	6th gear						Heading					Specification					--					
14.14	30 - 40 Km/h						sec		Target Balanced			14.36	--	5 29.16	14.80	10						
14.9	40 - 50 Km/h						sec					15.80	--	5 16.80	14.37	10						
14.15	50 - 60 Km/h						sec					18.80	--	5 18.80	16.30	10						
C	Engine Calibration Activities						Heading															
1.0	System Verification Trials						Heading															
1.1	Powertrain Characteristics (AC ON)						Target															
1.2.1	Powertrain Characteristics (AC OFF)						Target															
2.0	Vehicle Performance Trials						Heading															
2.1	Vehicle Speed detection						Target															
2.2	Engine Speed detection						Target															
2.3	Gear Detection						Target															
2.4	Driven Idle						Target															
3.0	Engine derate verification						Heading															
3.1	Limp home validation						Heading															
3.2	Reduced torque fault validation						Target															
4.0	Evaluation of Take-off ability, Acceleration and Deceleration						Pass															
D	Vehicle feature validation and Fuel Economy features / End of Line (EOL) parameters segregation and verification						Heading															
19	Vehicle Mass Estimation						Target															
20	Road Slope Estimation						Target															
20	Vehicle acceleration management (VAM tuning)						Target															
40	Electronic vehicle speed Limiter (EVSL) / Load based speed control (LBSC)						Target															
50	Vehicle speed limiter (Top and one-gear down)						Target															
60	EVSL+ VAM / LBSC+VAM						Target															
70	Gear Down Protection (GDP)						Target															
80	Idle shutdown						Target															
90	Average fuel economy (AFE)						Target															
100	IFE						Target															
100	Engine Brake control						Target															
120	Cruise control (CC)						Target															
330	Power Take off (PTO)						Target															
140	Remote PTO						Target															
150	Road Speed governor						Target															
160	Engine Protection shutdown						Target															
270	Powertrain protection						Target															
160	Trip info						Target															

Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification	Ref. Standard / Work Guideline	Simulation	Base Vehicle	Core Benchmark	Target	Test Vehicle	Status (RYG)	Remarks	CoC Remark								
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7																				
Annexure - 2																										
E	Environmental Trials Verification																									
1.0	Summer trials																	--								
1.4	Heading						--																			
1.4	Hot start								Target					--	Pass	Pass	NA	Carry over from OBD I validation								
1.5	Low idle governor calibration								Target					--	Pass	Pass	NA									
1.6	Driven idle								Target					--	Pass	Pass	NA									
1.7	Subjective Drivability Assessment								Target					--	Pass	Pass	NA									
2.0	Winter trials																									
2.2	Heading						--																			
2.2	Cold Start (at Ambient as per Application requirement)								Target					--	Pass	Pass	NA	Carry over from OBD I validation								
2.3	Smoke level (visually) during free acceleration after start								Target					--	Pass	Pass	NA									
2.4	Subjective Drivability Assessment								Target					--	Pass	Pass	NA									
6.0	High altitude trials/ calibration																									
6.2	Heading						--																			
6.2	Low idle governor calibration						gpm	New Development	Target					--	Pass	Pass	NA	Carry over from OBD I validation								
6.9.1	% Deterioration in Drivability with base calibration						%		Target					--	Pass	Pass	NA									
6.10.1	Turbocharger duty cycle mapping								Target					--	Pass	Pass	NA									
6.10.1.1	Smoke during driving present or not								Target					--	Pass	Pass	NA									
6.10.1.2	Assessment of turbocharger Whistling noise							New Development	Target					--	Pass	Pass	NA									
6.11.1	Power calculation						%		Target					--	Pass	Pass	NA									
F	Onboard Diagnostics																									
8.9	Heading						--																			
7.6	Derate Strategies for Critical Faults								Heading					--	Pass		NA	Carry over from OBD I validation								
8.9	Verification of OBD interface for Comprehensive Component Monitors								Heading					--	Pass		NA									
Sign Off															Index for Status											
Responsibility							Name						Signature					Off target, no resolution								
Prepared by: PAT Engineer							Debabrata Das/Swapnil Jadhav						 					Marginally off target, resolution planned/low risk								
Checked & Approved by: PAT Lead / Project Lead							Ajaykumar Jadhav / Sanjeev kumar						 					Met target, no concern								

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

PD Evaluation of Signa 2820.T BSVI RDE CNG

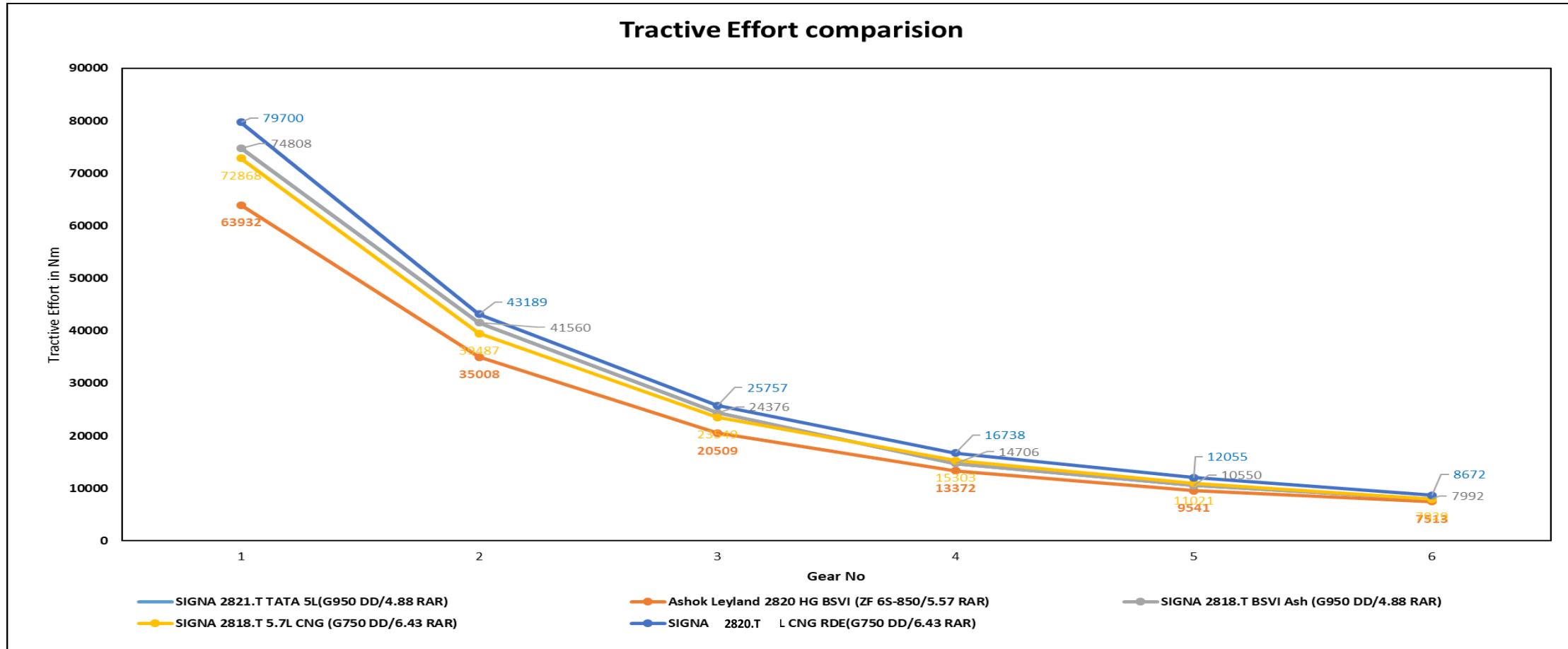
PED | Oct 23



Vehicle Specification Comparison

Sr. No.	Parameters	Ashok Leyland 2820HG	Signa 2818.T	Signa 2821.T	Signa 2818.T CNG	Signa 2820.T CNG
1	Engine	H-series 5.7L	Ashwamedh 5.6L	TATA 5L	TATA 697 5.7L SGI CNG OBD I	TATA 697 5.7L SGI CNG OBD II
1.1	No of cylinders	6	6	4	6	6
1.2	Engine Max. Power	200 HP @ 2400 rpm	180 HP @ 2300 rpm	197 HP @ 2200 rpm	176 HP @ 2250 rpm	204 HP @ 2250 rpm
1.3	Engine Max. Torque	700 Nm @ 1200-1900 rpm	850 Nm @ 1000 - 1600 rpm	850 Nm @ 1100 - 1600 rpm	650 Nm @ 1500 rpm	700 Nm @ 1500 rpm
1.4	Power to weight (HP/Ton)	7.14	6.43	7.04	6.29	7.28
1.5	Torque to weight (Nm/Ton)	25.00	30.36	30.36	23.21	25.00
1.6	Engine Calibration Number	--	Production Released	Production Released	57SGITCICBS6	--
1.7	Engine Calibration Maturity	100%	Production Released	Production Released	Develop Cal	Develop Cal
1.8	Emission Calibration Maturity	100%	100%	100%	Develop	Develop
1.9	Regeneration Strategy	SCR+HC	SCR	SCR	--	--
1.9.1	Exhaust Pipe Insulation	--	Fully wrapped metal insulation	Fully wrapped metal insulation	--	--
2	Fuel Economy Improvement Technologies					
2.1	Engine Cooling Fan	E-Viscous	E-Viscous (24" Fan)	E-Viscous (24" Fan)	E-Viscous (24" Fan)	E-Viscous (24" Fan)
2.2	Fuel Economy Switch	NA	3 mode (EBP)	3 mode (EBP)	2 mode (EP)	2 mode (EP)
3	Fuel Economy Improvement Technologies					
3.1	Fuel Tank type	Polymer	Polymer	Polymer	Metal Cylinder	Metal Cylinder
3.2	Driveline & Ratios	ZF 6S-850 DD, FGR- 8.51	G950 6S DD, FGR- 9.36	G950 6S DD, FGR- 9.36	G750 6S DD, FGR- 9.19	G750 6S DD, FGR- 9.19
3.3	Gear Ratio	1st: 8.51, 2nd: 4.66, 3rd: 2.73, 4th: 1.78, 5th: 1.27, 6th: 1, Rev:7.64	1st: 9.36, 2nd: 5.20, 3rd: 3.04, 4th: 1.83, 5th: 1.32, 6th: 1, Rev:8.36	1st: 9.36, 2nd: 5.20, 3rd: 3.04, 4th: 1.83, 5th: 1.32, 6th: 1, Rev:8.36	1st: 9.19, 2nd: 4.98, 3rd: 2.97, 4th: 1.93, 5th: 1.39, 6th: 1, Rev:8.36	1st: 9.19, 2nd: 4.98, 3rd: 2.97, 4th: 1.93, 5th: 1.39, 6th: 1, Rev:8.36
3.4	RAR	5.57	4.88	4.88	6.43	6.43
3.5	Tire size	295/90R20	295/90R20	295/90R20	295/90R20	295/90R20
4	EOL Parameters					
4.1	LBSC, VAM, GDP	NA	LBSC - 1900rpm, 1900rpm, 28000kg, VAM- 2.5kmph/Sec, 2.5kmph/Sec, 10kmph, 40kmph, GDP- 50kmph, 50kmph,	NA	NA	NA
4.2	RSG	80	80	80	74	74

Tractive Effort Comparison



- Signa 2820.T CNG RDE is better as compared to Signa 2818.T BSVI, Signa 2828.T CNG and AL 2820.T BSVI

Min & Max Speed

	Ashok Leyland 2820 HG BSVI Eaton 6S DD_RAR 5.57		Signal 2818.T BSVI Ash G950 6S DD_RAR 4.88		Signal 2821.T 5L BSVI G950 6S DD_RAR 4.88		Signal 2818.T BSVI SGI CNG G750 6S DD_RAR 6.43		Signal 2820.T SGI CNG RDE G750 6S DD_RAR 6.43	
Mode	--		HM		HM		HM		HM	
Gear no	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
1st gear	2.7	12.4	3.2	10.9	2.9	10.9	2.1	9.7	2.4	9.7
2nd gear	5.3	21.1	5.8	19.8	4.7	19.0	4.2	16.7	4.5	16.5
3rd gear	8.6	34.7	10.1	33.7	9.2	32.0	6.7	27.2	7.6	27
4th gear	12.8	51.7	16.4	48.3	15.4	52.1	10.2	41.2	11.5	40.7
5th gear	18.9	71.4 (2600 rpm)	21.9	55.0 (1850 rpm)	21.2	70.8 (2350 rpm)	15.4	55.9 (2550 rpm)	15.2	54.6
6th gear	22.5	78 (2200 rpm)	29.4	60.0 (1500 rpm)	27.4	76.5 (1900 rpm)	19.1	74.8 (2300 rpm)	21.3	75

Performance Evaluation Summary

Signa 2820.T CNG RDE Performance Evaluation													
Parameter	Unit	Target 2820.T CNG BSVI RDE	PALS Position	AL 2820 HG BSVI RAR 5.57	SIGNA 2818.T BSVI Ashwamedh RAR 4.88 HM	SIGNA 2821.T 5L BSVI RAR - 4.88 HM	SIGNA 2818.T 5.7L CNG BSVI RAR - 6.43 HM	SIGNA 2818.T 5.7L CNG RAR 6.43 LM	SIGNA 2818.T 5.7L CNG RAR 6.43 HM	Margin wrt target	Margin wrt Base	PALS Status	Remarks
Accleration at Rated Load													
0 - 60 kmph	sec	≤ 65.26	UC	50.43	49.38	50.83	65.26	86.18 (0 - 55 Kmph)	54.23	20.34%	20.34%		
0 - 1000 m	sec	≤ 83.94	UC	75.84	74.89	76.50	83.94	90.31	77.12	8.84%	8.84%		
Driveability(Rate of acceleration) at Rated Load													
1st Gear (4-6 Kmph)	sec	Not Applicable		0.84	1.02	1.02	0.65	0.89	0.82	Not Applicable	Not Applicable	Not Applicable	
2nd Gear (8-12 Kmph)	sec			2.32	1.92	1.99	1.35	1.55	1.47				
3rd Gear (12 - 20 Kmph)	sec			6.04	4.86	5.04	3.83	4.45	3.78				
4th Gear (19 - 31 Kmph)	sec			9.95	10.18	11.47	9.52	12.01	7.92				
5th Gear (25 - 55 Kmph)	sec			37.68	20.36(25 - 50 kmph)	37.70	42.02	61.71	35.28				
6th Gear (36 - 60 kmph)	sec			41.58	37.76	41.90	47.89	51.0(36 - 54)	35.60				
1st Gear (4-6 Kmph)	kmph/sec	Not Applicable		2.38	1.96	1.96	3.08	2.25	2.44	Not Applicable	Not Applicable		
2nd Gear (8-12 Kmph)	kmph/sec			1.72	2.08	2.01	2.96	2.58	2.72				
3rd Gear (12 - 20 Kmph)	kmph/sec			1.32	1.65	1.59	7.83	1.80	2.12				
4th Gear (19 - 31 Kmph)	kmph/sec			1.21	1.18	1.05	3.15	1.00	1.52				
5th Gear (25 - 55 Kmph)	kmph/sec			0.80	1.23	0.80	0.71	0.49	0.85				
6th Gear (36 - 60 kmph)	kmph/sec			0.58	0.64	0.57	0.50	0.35	0.67				
Bath-Tub Driveability													
IIIrd Gear													
15 - 20 Kmph	Sec	≤ 3.20	UC	3.05	3.06	3.24	2.46	2.88	2.11	34.06%	14.11%		
20 - 25 Kmph	Sec	≤ 2.47	UC	2.36	2.85	2.48	2.45	3.33	2.46	0.40%	-0.47%		
25 - 30 Kmph	Sec	--	UC	--	3.21	2.62	--	--	--				
IV th Gear													
20 - 25 Kmph	Sec	≤ 5.94	UC	5.66	5.31	4.55	3.25	4.36	3.19	46.30%	1.74%		
25 - 30 Kmph	Sec	≤ 4.17	UC	3.97	4.19	4.33	3.61	5.41	3.29	21.10%	8.77%		
30 - 35 Kmph	Sec	≤ 3.70	UC	3.42	4.20	3.89	3.70	5.75	3.60	2.70%	2.64%		
35 - 40 Kmph	Sec	≤ 5.52		3.61	--	3.82	5.52	6.01	6.01	-8.88%	-8.91%		
V th Gear													
30 - 40 Kmph	Sec	≤ 12.11	UC	11.53	11.97	12.83	10.13	18.71	9.73	19.65%	3.99%		
40 - 50 Kmph	Sec	≤ 13.83	UC	9.855	12.24	12.74	13.83	21.17	10.93	20.97%	20.98%		
50 - 60 Kmph	Sec	--	UC	--	13.72	14.14	--	--	--				
VI th Gear													
30 - 40 Kmph	Sec	≤ 29.15	UC	27.76		17.44	24.36	22.72	14.80	49.23%	39.24%		
40 - 50 Kmph	Sec	≤ 16.58	UC	15.79	16.85	17.68	15.89	27.13	14.37	13.33%	9.58%		
50 - 60 Kmph	Sec	≤ 18.80	UC	14.89	16.87	17.77	18.80	--	16.30	15.34%	13.30%		
60 - 70 Kmph	Sec	--	UC	--	18.25	20.12	31.58	--	--				
Restart and Sustain Gradeability													
Maximum (sustained) Gradeability (forward)	%	≥ 26.38	UC	--	27.77	--	27.58	--	29.51	11.87%	7.00%		
Max Restart Gradeability (forward)	%	≥ 18.00	UC	--	23.04	--	21.6	--	23.11	28.40%	7.00%		

Sigma 2820.T CNG RDE Cx BSVI RDE Performance. PALS position achieved “L” against “UC”

Acceleration

❑ Acceleration of Sigma 2820.T CNG RDE was found to be better as compared to target in 0-60 kmph speed and 0-1000 meter band

Bath tub Drivability

❑ Drivability of Sigma 2820.T CNG RDE was found to be better as compared to target in 3rd ,4th ,5th and 6th gear.

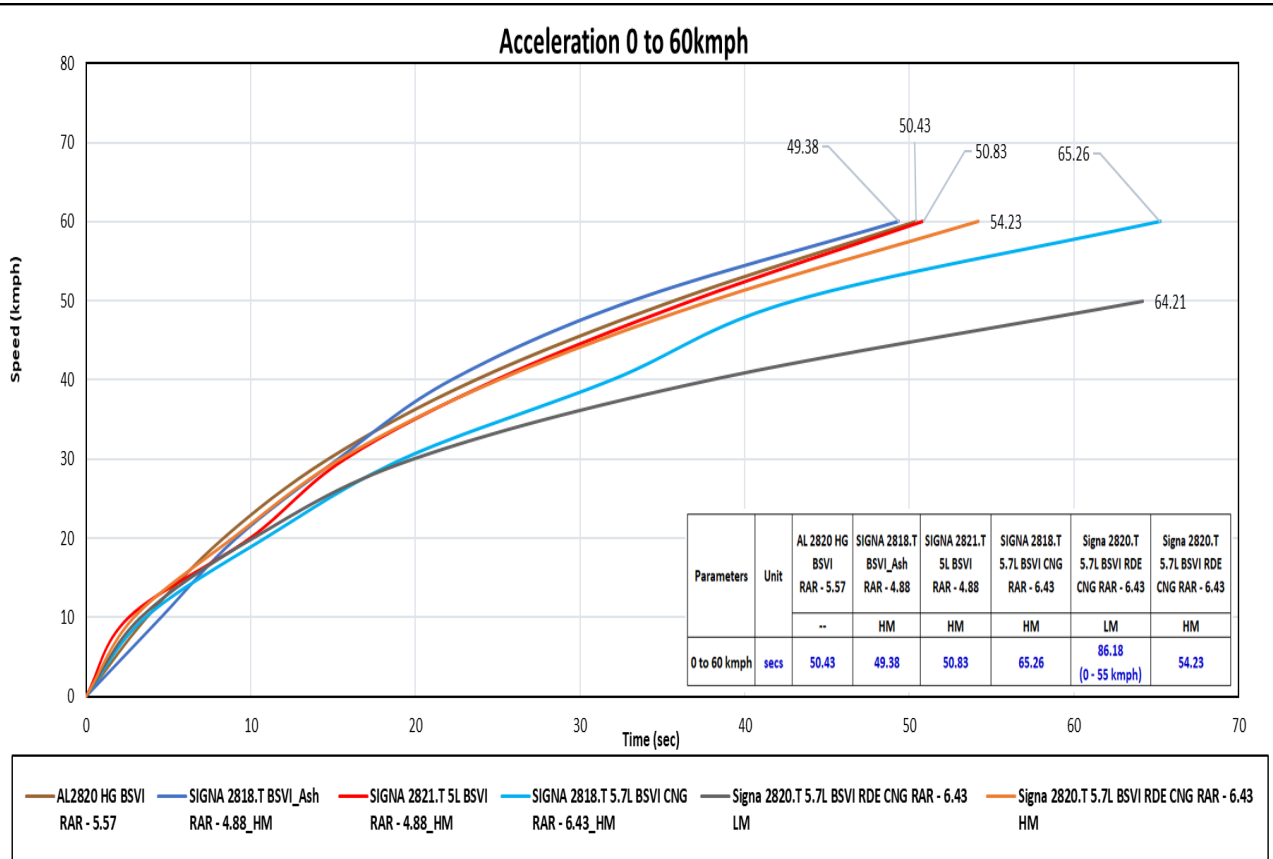
❑ Drivability of Sigma 2820.T CNG RDE is not meeting target in 4th Gear (35 – 40 Kmph)

Sustain & Restart Gradeability

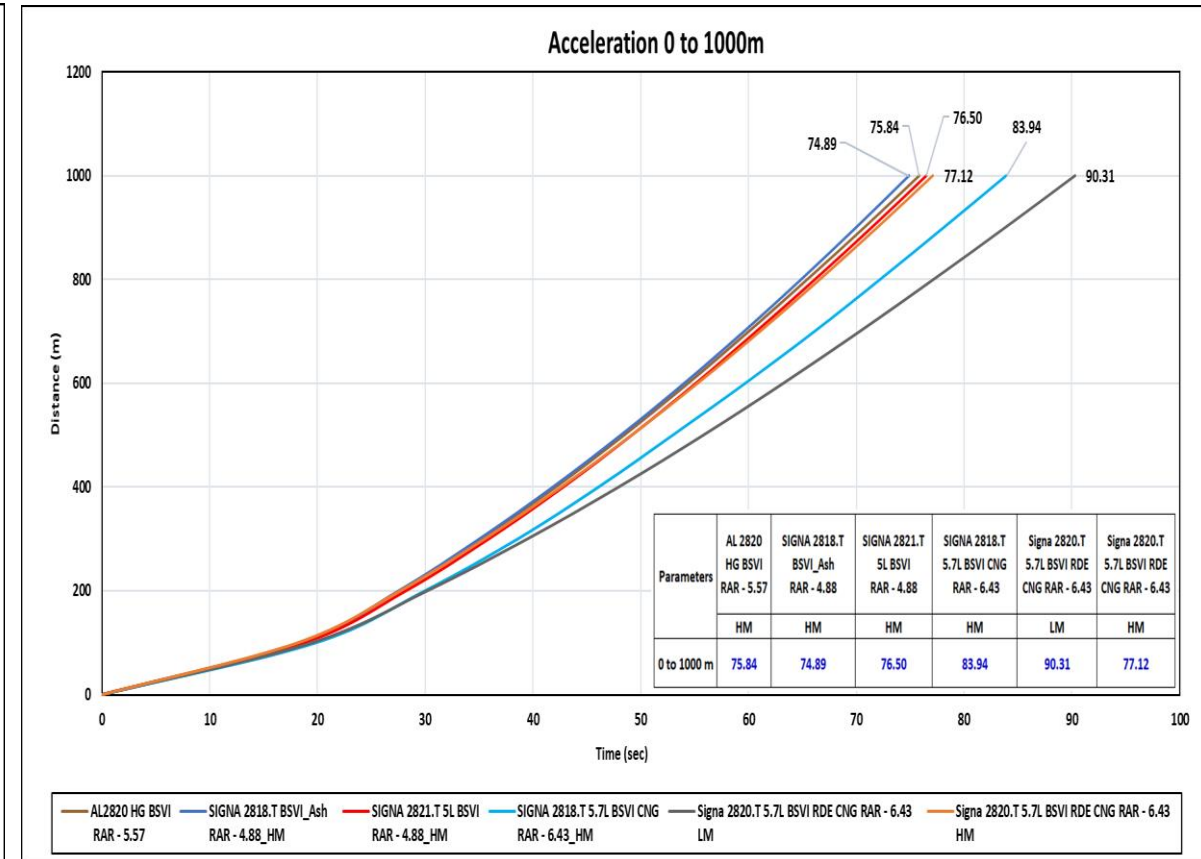
Sustain and Restart Gradeability is Extrapolated from Sigma 2818.T CNG BSVI as the driveline is same & there is 50Nm torque increase in CNG vehicle

Acceleration

0- 60kmph



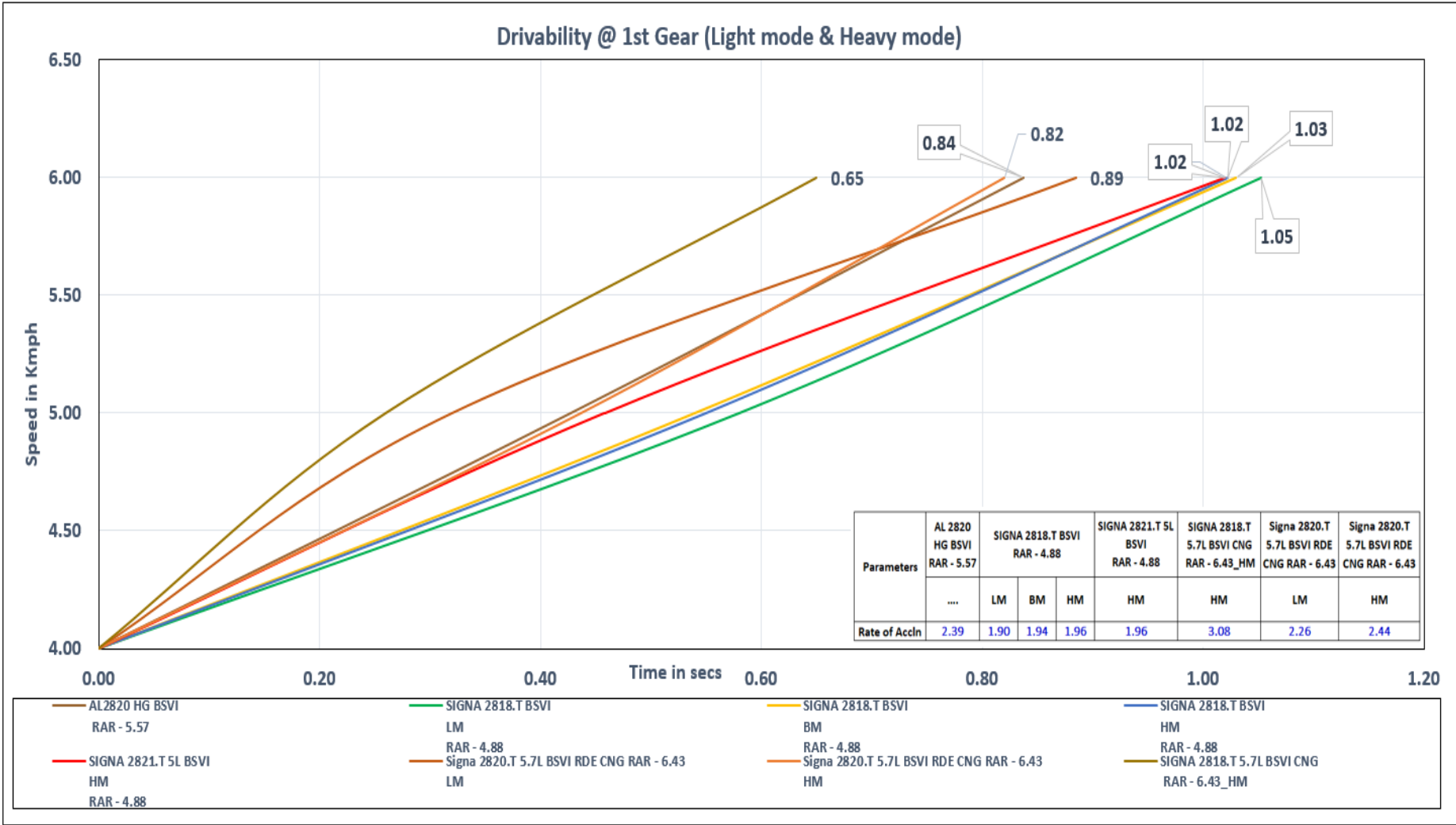
0- 1000meter



Remarks:

- Signa 2820.T BSVI RDE CNG vehicle 0-60kmph & 0-1000m **is better as compared** to Signa 2818.T BSVI CNG.

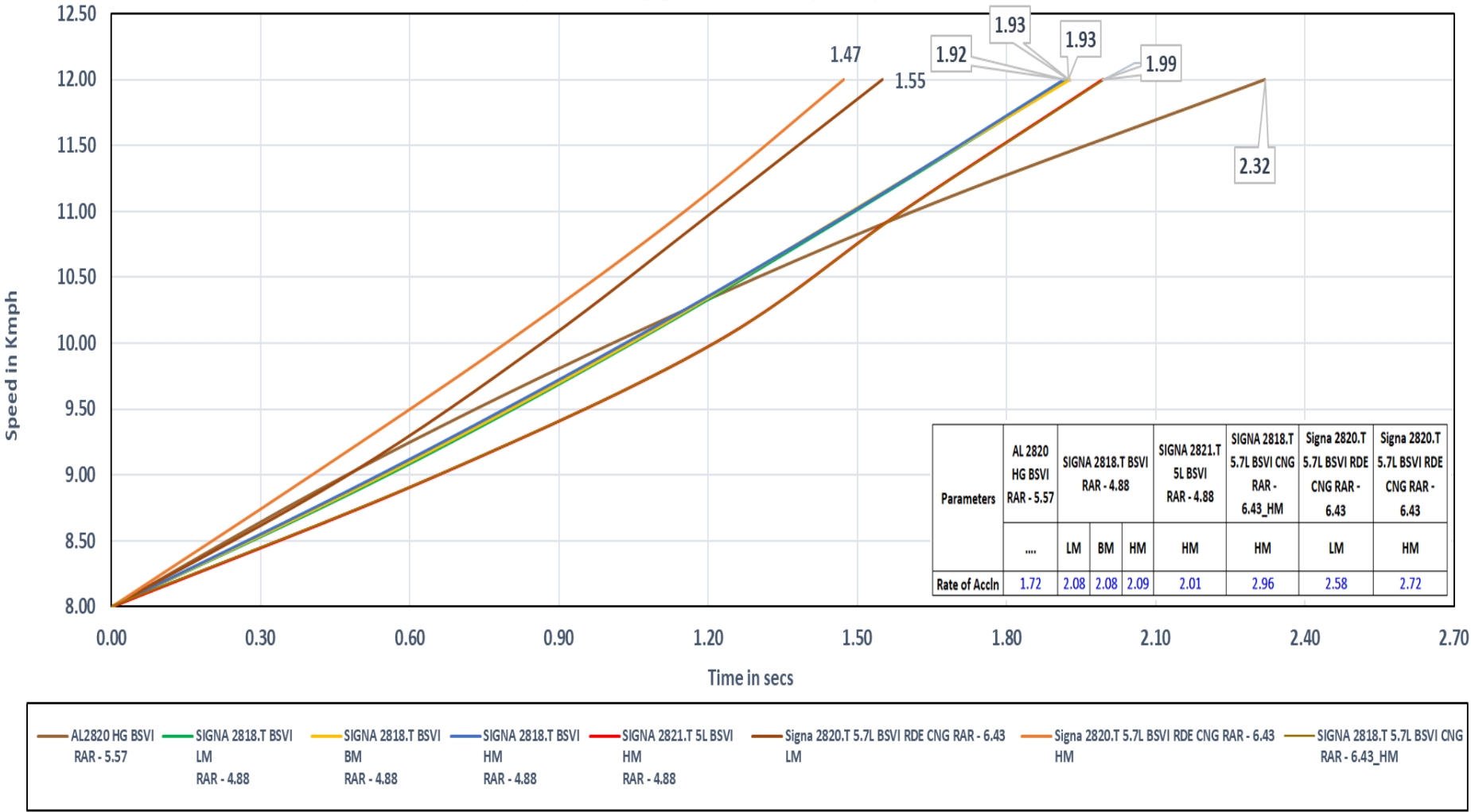
1st Gear Wide Open Throttle Drivability



Remarks: Signa 2820.T CNG BSVI RDE WOT performance is inferior as compared to Signa 2818.T CNG BSVI in 1st Gear.

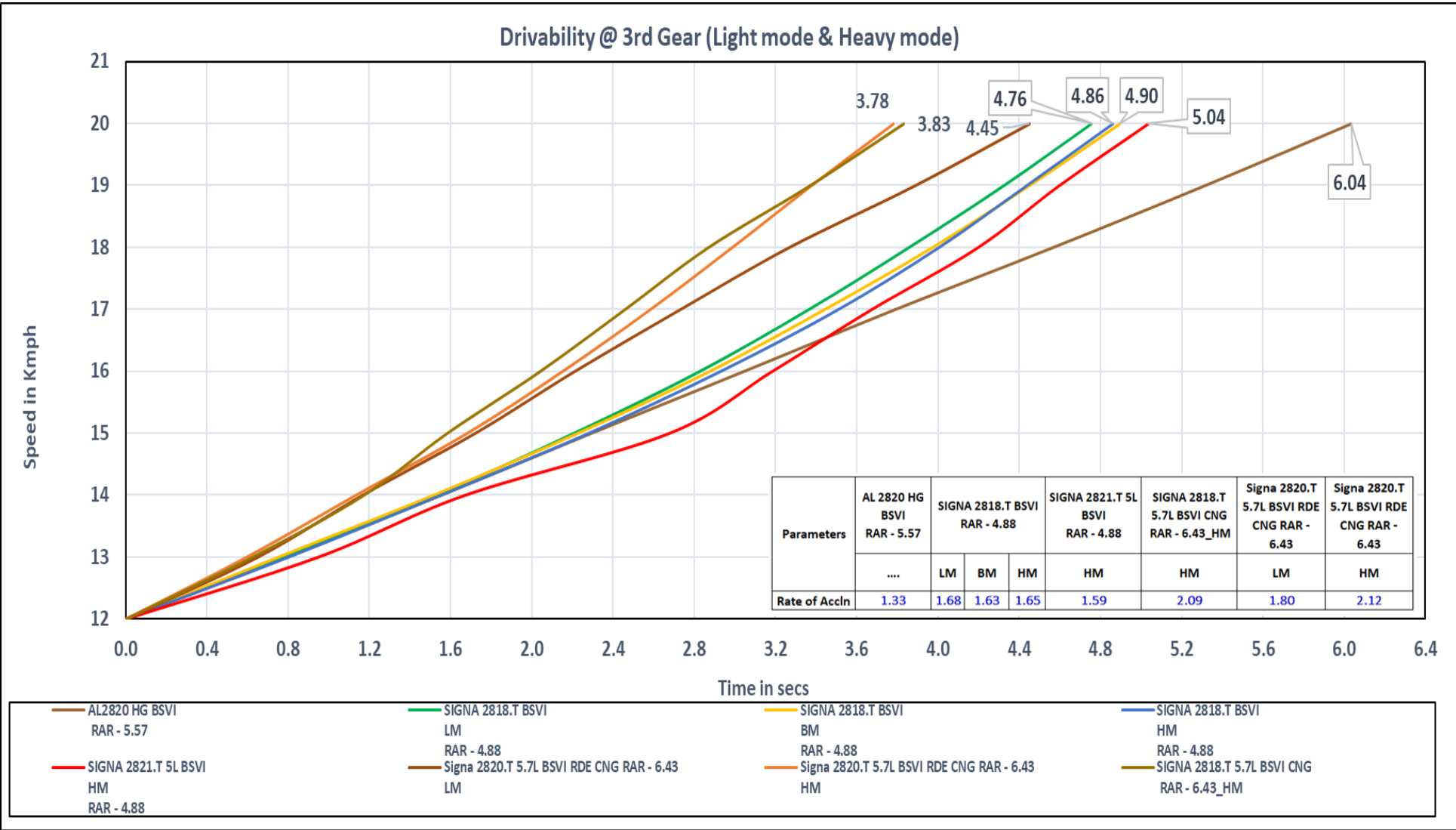
2nd Gear Wide Open Throttle Drivability

Drivability @ 2nd Gear (Light mode & Heavy mode)



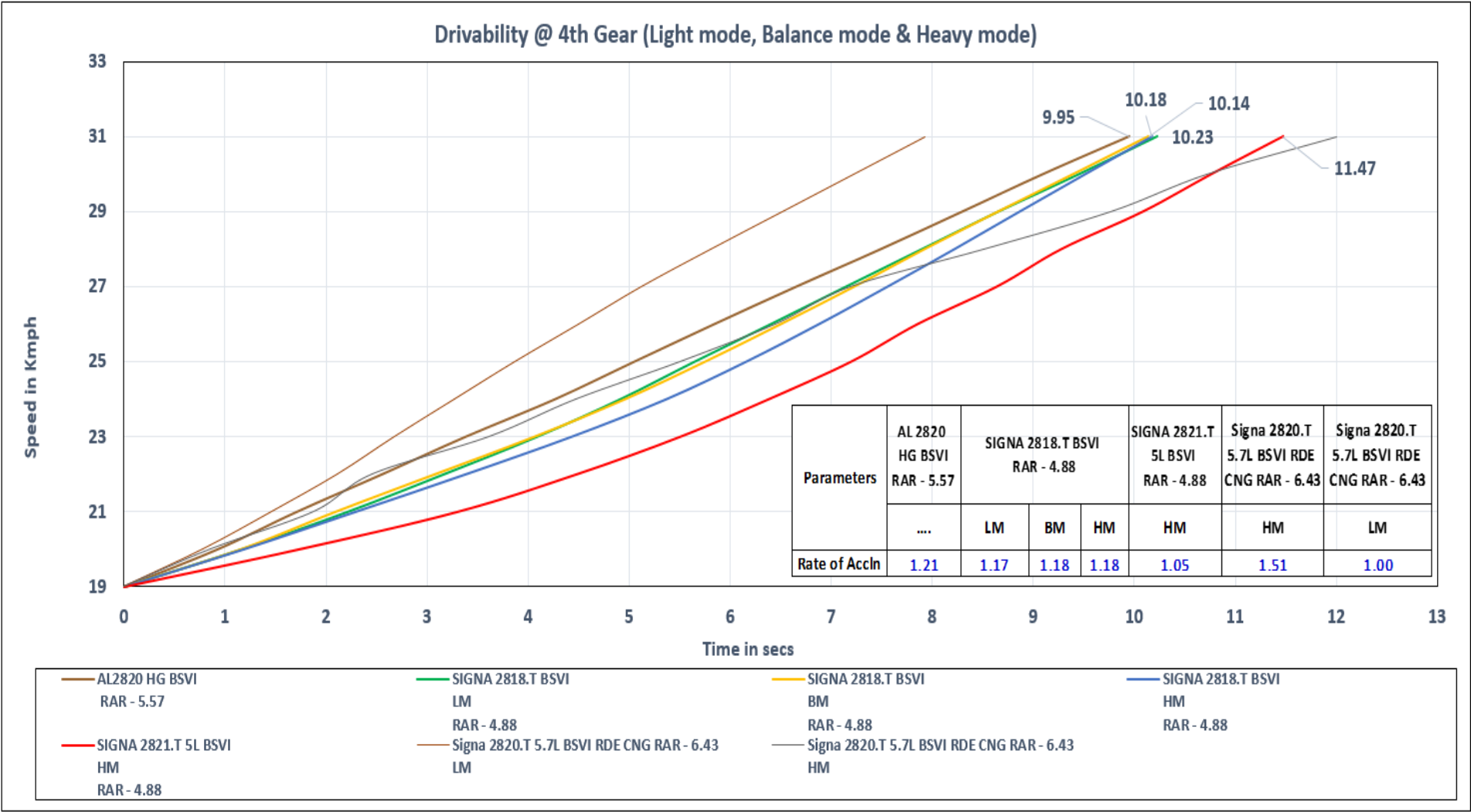
Remarks: Signa 2820.T CNG BSVI RDE WOT performance is inferior as compared to Signa 2818.T CNG BSVI in 2nd Gear.

3rd Gear Wide Open Throttle Drivability



Remarks: Signa 2820.T CNG BSVI RDE WOT performance is better as compared to Signa 2818.T CNG BSVI in 3rd Gear.

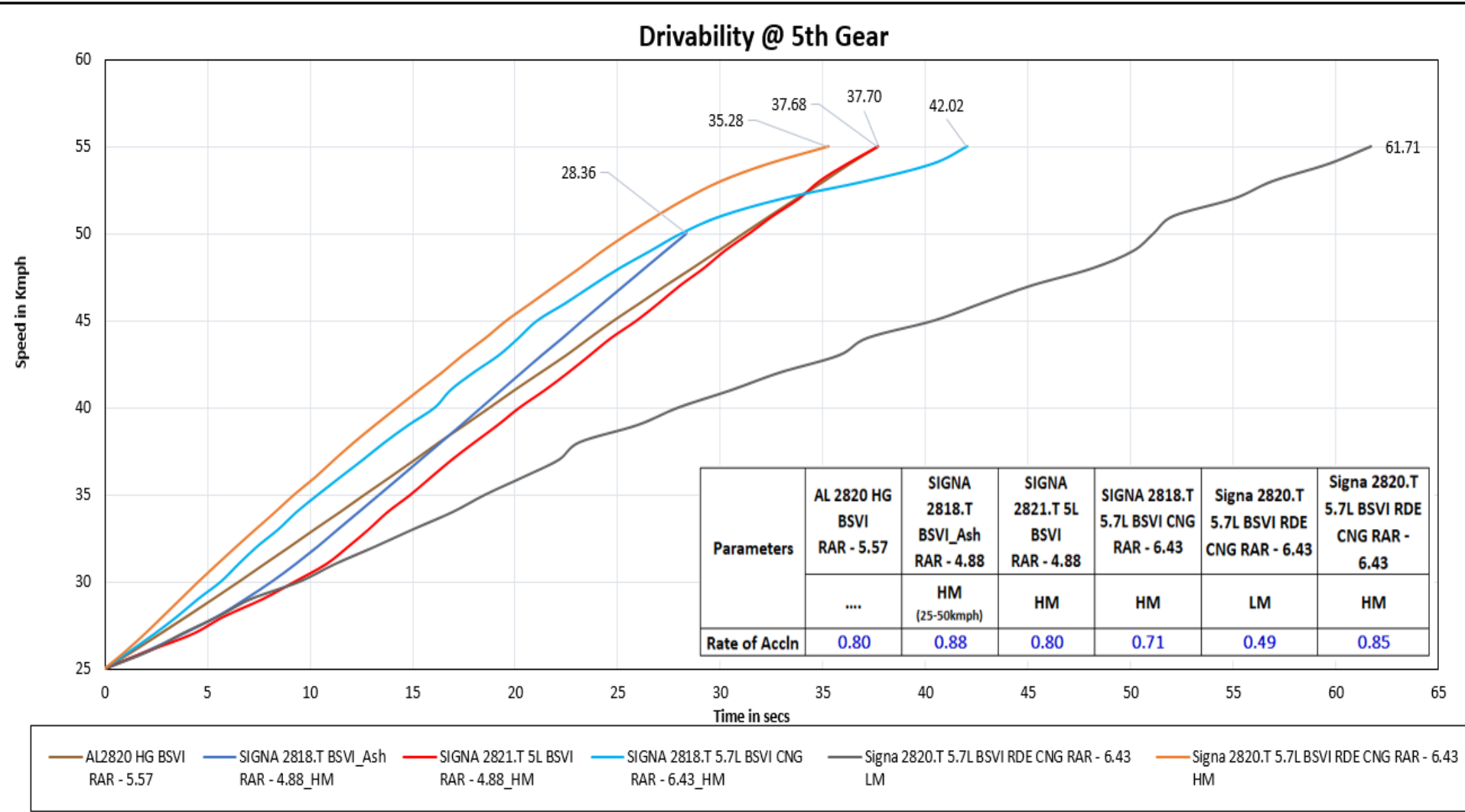
4th Gear Wide Open Throttle Drivability



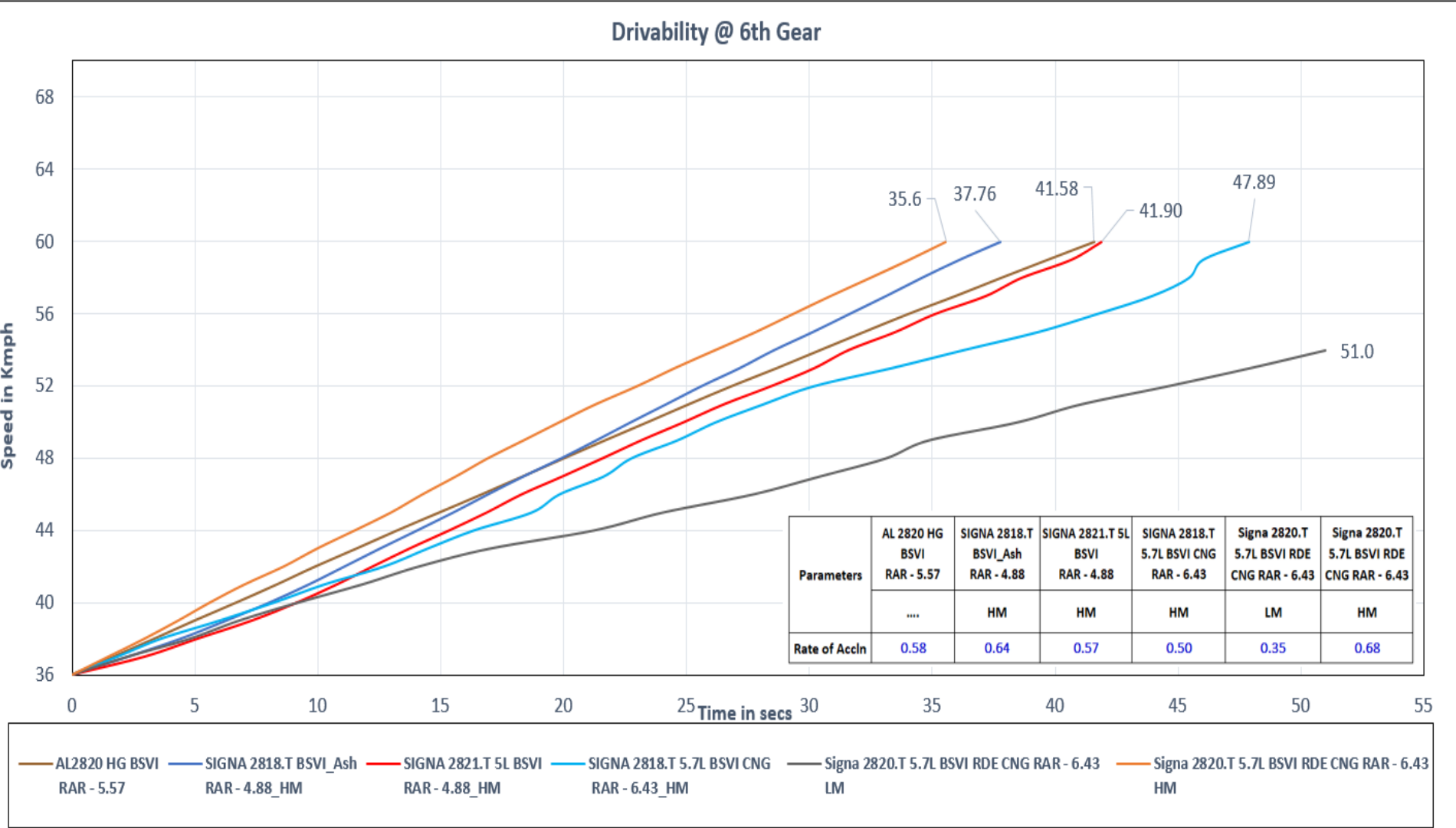
Remarks: Signa 2820.T CNG BSVI RDE WOT performance is better as compared to Signa 2818.T CNG BSVI in 4th Gear.

5th Gear Wide Open Throttle Drivability

Remarks: Signa 2820.T CNG BSVI RDE WOT performance is better as compared to Signa 2818.T CNG BSVI in 5th Gear.



6th Gear Wide Open Throttle Drivability



Remarks:

- Signa 2820.T CNG BSVI RDE WOT performance is better as compared to all vehicles in top gear.

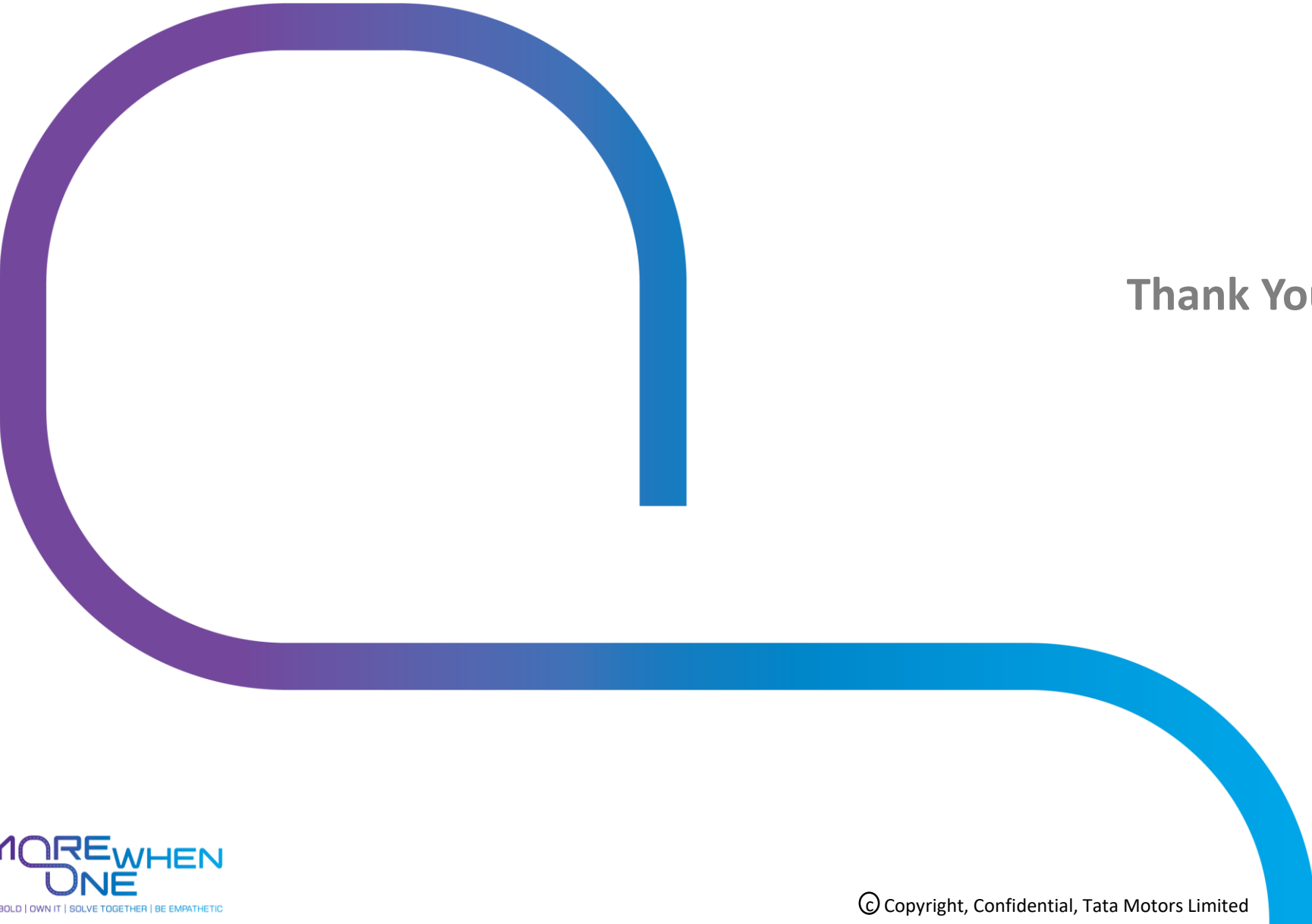
Bathtub Drivability



Remarks:

- Signa 2820.T CNG RDE BSVI is better than Signa 2818.T CNG BSVI in all gears all speed band except 4th Gear 35 – 40 kmph.





Thank You

TEST REQUISITION

Ref.	NA	Date:	06/06/2023
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 2820.T BSVI CNG				
2	Part Number	51832155R,				
3	Drawing number	518300000014,	Mod status	D,2,	Mod Date	25/10/2023
4	Vehicle Model / Project	Signa 2820.T CNG RDe BSVI				
5	WBS Number	P.22.EA.497 -MHCV 5.7 SGI Truck BS6 RDE for 28T Truck				
6	Supplier	Tata Motrs				
7	Number of samples					
8	Inspection report	Enclosed	☐ Yes	☒ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☒ No	If No commitment date:	
10	Suppliers test report	Enclosed	☒ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐		Weight Reduction	☐
		Alternate source development	☐		Cost Reduction	☐
		Data generation Test	☐		Regulatory test	☐
		Any other (pl. specify)	☐		New Development	☒
12	What testing required?					
13	Remarks	PD testing as per DVP signoff.				
14	Peripheral Parts Availability	NO				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN	DATE: 30/10/2023			
	VEHICLE TESTING DEPARTMENT				
Description: Signa 2820.T CNG RDe BSVI		Test Item No: 43364			
		New	☒	Moderate	☒
		Major	☒	Minor	☒

Part No: 51832155R,		Modification No: D,2,	Enclosures:
Drawing No: 518300000014,			
Make: Tata Motrs		WBS Number: P.22.EA.497 -MHCV 5.7 SGI Truck BS6 RDE for 28T Truck	Copies to :
Vehicle Type: Signa 2820.T CNG RDe BSVI			
Test Requested by:		M & HCV	Equipments:
Test Requisition Reference:		Testing-Outdoor/15522	
Reason for Test/s:		New development	
Regulatory Requirement:		a) Component: b) Vehicle:	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	PD trials	As per PED	As per PALS	1	Debabrata Das	Beta	Nil

REMARKS: Test plan for DVP execution	INTERNAL REF:	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	CHECKED BY	APPROVED BY